

2. RESEARCH METHODOLOGY

2.1 RESEARCH PHILOSOPHY

The research philosophy used in this study is positivist. The knowledge is gained from observable and measurable phenomena, which is appropriate because there are objective, quantifiable answers to the research questions (F1 scores, Likert means, drift degradation). A deductive approach was followed – questions were based on existing theory and then tested with the empirical data. This is in line with the scientific method outlined by Stallings (2023) and is in line with the Pearson BTEC D1 criterion for philosophical justification.

2.2 RESEARCH DESIGN CONCURRENT MIXED METHODS

A concurrent mixed-methods design was used, including three independent strands. This triangulation method helps to ensure that no one method has all the weight of the evidence and that findings converge across the qualitative, quantitative, and experimental strands.

Method	Purpose	Participants / Data	Analysis
Quantitative Survey	Practitioner attitudes on AI, XAI, FL, adversarial risk	n=22 CI/cybersecurity professionals	Descriptive stats; Likert analysis; frequency distributions
Algorithm Simulation	IDS performance benchmarking	SWaT + CIC-IDS datasets	Precision/Recall/F1/Accuracy/Latency
Literature Review	Secondary evidence synthesis	10 peer-reviewed / institutional sources	Evidence matrix; gap analysis

Method	Purpose	Participants / Data	Analysis
Concept Drift Analysis	Model degradation over time	30-week simulated ICS deployment	F1 trend; delta calculations
Adversarial Testing	Robustness evaluation	10 attack types; 500 samples each	Evasion rate; composite risk score

2.3 PRIMARY DATA COLLECTION SURVEY (N=22)

The survey was designed in Google Forms and sent out through LinkedIn, IISP forums, and direct contacts for a period of 21 days. It included 15 questions: 7 Likert scale (1-5) questions, 5 categorical/multiple choice questions, 1 multi-select barriers question, and 1 optional open text question. It was a purposive sampling with a final sample of n=22 valid responses from eight industry sectors. 72.7% possessed 0-2 years of experience – this is an admitted limitation, as discussed in Sections 6.1 and 8.3. Ethical compliance: 100% anonymity, UK GDPR, digital consent, UK based data storage.

2.4 SECONDARY DATA COLLECTION

A systematic literature review was performed from IEEE Xplore, ACM Digital Library, Google Scholar, and NCSC publications. Published within the last 5 years, peer-reviewed or institutional authority, directly relevant to at least one RQ. Ten sources were selected.

2.5 ETHICS AND COST

Method	Cost	Access	Ethics	Sample
Primary Survey	£0	LinkedIn / IISP / direct contacts	Anonymous; GDPR; consent	n=22
Algorithm Simulation	~£220	Air-gapped digital twin	NCSC disclosure; payloads contained	SWaT + CIC-IDS
Literature Review	£0 (TUF)	IEEE/ACM/Scholar	Harvard referencing	10 sources

Method	Cost	Access	Ethics	Sample
Adversarial Testing	~£80	Air-gapped digital twin	Payloads contained	10 types, n=500